

Mathematics for Biomedical Engineering

L6:

Applications of Integration

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- To understand how to use the interpretation of the definite integral as the limit of a sum for calculating geometric and physical quantities;
- To be able to calculate volumes of revolution, lengths of curves, and areas of surfaces of revolution;
- To be able to calculate centres of mass, moments of inertia, mean and root-mean-square values.

Fundamental Theorem of Calculus

$$\lim_{\delta x \rightarrow 0} \sum_{x=a}^{x=b} y(x) \delta x = \int_a^b y(x) dx$$

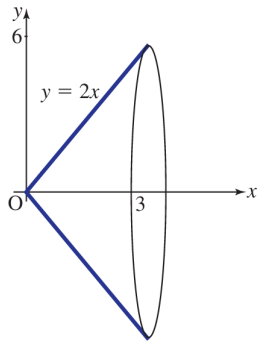
Given that integration is a process of adding or 'summation', it is possible to apply integration to finding areas, volumes, centres of gravity and many other physical quantities.

- 1 Volumes of Revolution
- 2 Calculating Centres of Mass
- 3 Moment of Inertia
- 4 Length of a Curve
- 5 Area of a Surface of Revolution
- 6 Mean Value and Root-Mean-Square Value

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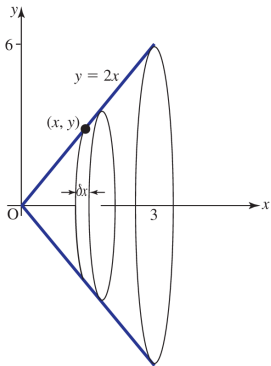
Volumes Generated by Rotating Areas about the x Axis

We start by studying how to find volumes of solids formed when areas are rotated around the x or y axes.



- Rotating the area under $y = 2x$ a full revolution around the x axis forms a cone;
- Such a three-dimensional body is known as a **solid of revolution** or a **volume of revolution**.

Volumes Generated by Rotating Areas about the x Axis



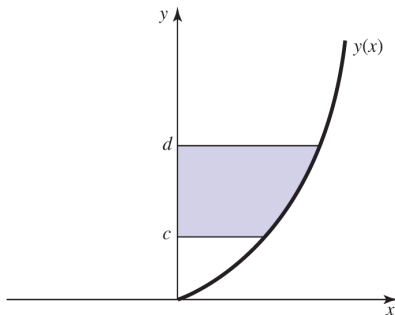
- We can find the volume of this solid by assuming that it is composed of lots of **thin circular discs** all aligned perpendicular to the x axis;
- Each disc has radius $y = 2x$ and thickness δx ;
- The volume of a disc is $\pi y^2 \delta x = 4\pi x^2 \delta x$;
- The **total volume of the solid** is

$$\lim_{\delta x \rightarrow 0} \sum_{x=0}^{x=3} 4\pi x^2 \delta x = \int_0^3 4\pi x^2 dx = 36\pi$$

General Calculation

If the area under the graph of $y(x)$ between $x = a$ and $x = b$ is rotated about the x axis, the volume of the solid formed is

$$\lim_{\delta x \rightarrow 0} \sum_{x=a}^{x=b} \pi y^2 \delta x = \int_a^b \pi y^2 dx$$

Volumes Generated by Rotating Areas about the y Axis

General Calculation

If the area between the graph of $y(x)$ and the y axis between $y = c$ and $y = d$ is rotated about the y axis, the volume of the solid formed is

$$\lim_{\delta y \rightarrow 0} \sum_{y=c}^{y=d} \pi x^2 \delta y = \int_c^d \pi x^2 dy$$

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Calculating Centres of Mass

The **centre of mass** is the point where, for many purposes, all the mass can be assumed to be located.

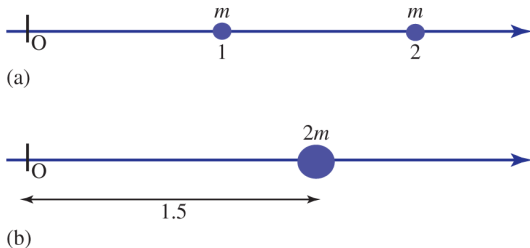
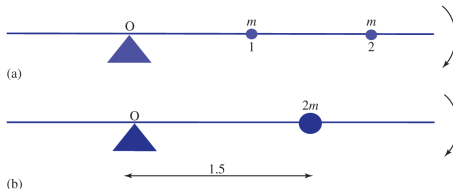


Figure: (b) The location of the centre of mass of the objects in (a).

It can be thought of the point where we could imagine placing a pivot to achieve a **perfectly balanced system**.

Moment

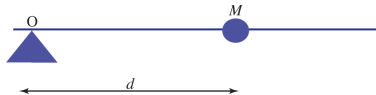
It can also be thought as follows: If a pivot is placed at the origin then the two masses at $x = 1$ and $x = 2$ together (a) have the same **turning effect** or **moment** as a single mass $2m$ located at $x = 1.5$ (b).



Definition of Moment

Given a mass M located a distance d from O , its moment about O is defined to be

$$\text{moment} = Md$$



The larger the moment, the larger the turning effect.

Calculation of the Centre of Mass Using Moments

Key Point

For a collection of n masses, $m_1, m_2, \dots, m_n$, the moment of the total mass, $\sum_{i=1}^n m_i$, located at the centre of mass, $\bar{x}$, is equal to the sum of the moments of the individual masses, $m_i x_i$. Thus,

$$\bar{x} \sum_{i=1}^n m_i = \sum_{i=1}^n m_i x_i$$

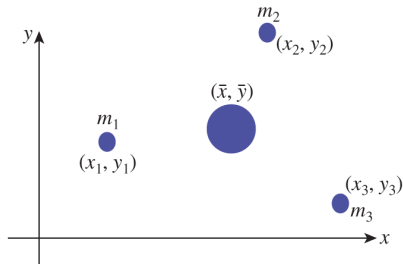
Solving for $\bar{x}$ we have:

x-component of the centre of mass

$$\bar{x} = \frac{\sum_{i=1}^n m_i x_i}{\sum_{i=1}^n m_i}$$

Centre of Mass in the xy Plane

When n masses are distributed anywhere in the xy plane, the centre of mass has a y -coordinate, $\bar{y}$, in addition to the x -coordinate, $\bar{x}$.



x - and y -components of the centre of mass

The centre of mass of n masses, $m_1, m_2, \dots, m_n$, located at $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ has coordinates

$$\bar{x} = \frac{\sum_{i=1}^n m_i x_i}{\sum_{i=1}^n m_i}, \quad \bar{y} = \frac{\sum_{i=1}^n m_i y_i}{\sum_{i=1}^n m_i}$$

Finding the Centre of Mass of a Plane, Uniform Lamina

A **lamina** is a thin sheet of material, such as a plane sheet of metal.

Its mass, M , is not located at individual points but is **distributed continuously** throughout the lamina. We will assume that the mass is **distributed uniformly**.

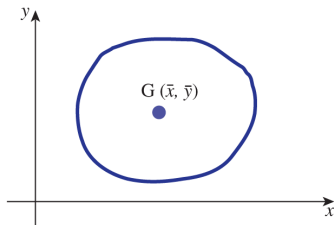


Figure: The centre of mass of the lamina is located at $G(\bar{x}, \bar{y})$.

We can find the position of the centre of mass G by:

- ➊ Dividing the lamina into many small pieces;
- ➋ Finding the mass of each piece;
- ➌ Calculating the moment of each piece about the axes;
- ➍ The sum of these moments about the y axis will be equal to $M\bar{x}$. The sum of these moments about the x axis will be equal to $M\bar{y}$.

Mass per Unit Area

Suppose that we have a uniform lamina and suppose we select a piece of the lamina that has area equal to 1 unit.

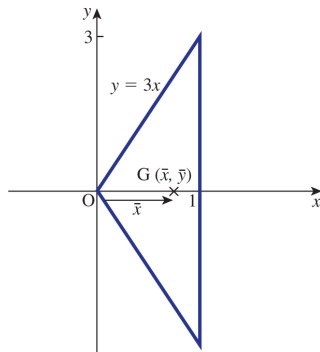
We define m as the mass of such a piece and we call it the **mass per unit area**.

The mass of any other piece can be expressed in terms of m ; e.g. an area of 2 units must have a mass $2m$, an area of 3 units must have a mass $3m$.

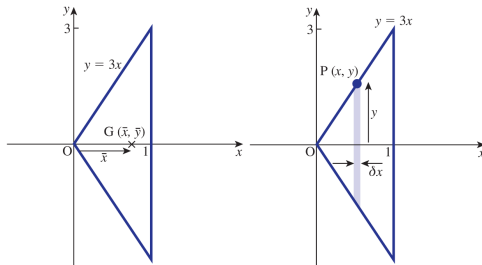
Key point

If a lamina has mass per unit area m , then the mass of part of the lamina having area A is mA .

Example: Calculation of Centre of Mass of a Lamina



Example: Calculation of Centre of Mass of a Lamina



By symmetry the centre of mass lies on the x axis. Thus, $\bar{y} = 0$. To find $\bar{x}$:

- ① Divide the lamina into small pieces: thin vertical slices in this case (see figure, right);
- ② Find the mass of each piece: $m(2y\delta x)$;
- ③ Calculate the moment of each piece about the y axis: $mx(2y\delta x)$;
- ④ The sum of these moments about the y axis will be equal to $M\bar{x}$:

$$\int_0^1 2mxy \, dx = M\bar{x}, \quad \text{with } M = 3m$$

Solution: $\bar{x} = \frac{2}{3}$. Thus, the coordinates of the centre of mass are $(\frac{2}{3}, 0)$. Note that $\bar{y} = 0$ due to the symmetry of the problem.

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Moment of Inertia of a Lamina

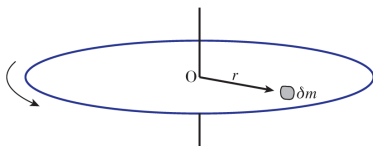
The **moment of inertia** or **rotational inertia** of an object about an axis is a measure of the resistance to rotation of the object.

It is essential for understanding **rotational dynamics** of solid bodies.

It plays the same role when dealing with rotating bodies as the mass of an object plays when dealing with motion in a line:

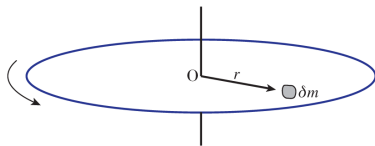
- An object with large mass needs a large force to achieve a given acceleration;
- An object with large moment of inertia needs a large turning force to achieve a given angular acceleration.

We can calculate the moment of inertia of a lamina by **dividing it into a large number of small pieces or elements**.



Moment of Inertia of a Lamina

- The moment of inertia of the small element is defined to be $r^2 \delta m$;
- The total moment of inertia of the lamina is $\sum r^2 \delta m$;
- As $\delta m \rightarrow 0$, moment of inertia $I = \int r^2 dm$.



Definition of Moment of Inertia

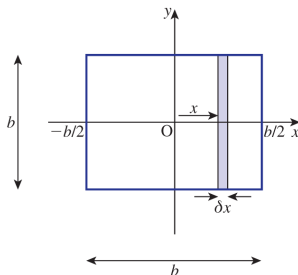
$$\text{moment of inertia } I = \int r^2 dm$$

where the limits of integration are chosen so that the entire lamina is included.

The SI unit of moment of inertia is Kg m^2 .

Example: Calculation of Moment of Inertia of a Square

Calculate the moment of inertia about the y axis of the square lamina below of mass M and width b . Let the mass per unit area of the lamina be m .



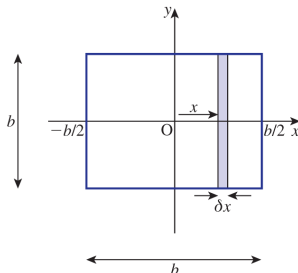
Steps:

- ① Divide the lamina into many small pieces: thin vertical strips in this case;
- ② Calculate the moment of inertia of each strip;
- ③ Add contributions from all strips;
- ④ Calculate I as the sum as $\delta x \rightarrow 0$;
- ⑤ Express I in terms of the total mass M ;

Important: The limits on the integral must be chosen so that the whole lamina is included.

Example: Calculation of Moment of Inertia of a Square

Calculate the moment of inertia about the y axis of the square lamina below of mass M and width b . Let the mass per unit area of the lamina be m .



Steps:

- ① Divide the lamina into many small pieces: thin vertical strips in this case;
- ② Calculate the moment of inertia of each strip: $x^2 mb \delta x$;
- ③ Add contributions from all strips: $\sum_{x=-b/2}^{x=b/2} x^2 mb \delta x$;
- ④ Calculate I as the sum as $\delta x \rightarrow 0$: $I = \int_{-b/2}^{b/2} x^2 mb dx = \frac{mb^4}{12}$;
- ⑤ Express I in terms of the total mass M : $M = mb^2$, $I = \frac{Mb^2}{12}$.

Important: The limits on the integral must be chosen so that the whole lamina is included.

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Length of a Curve

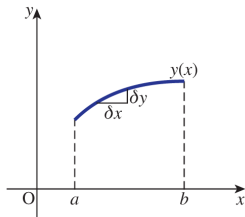
Given a curve with equation $y = f(x)$, then the length of the curve between the points $x = a$ and $x = b$ is given by the formula

Formula

$$\text{length of curve} = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

We can prove this formula by:

- 1 using Pythagora's theorem to find the length of the hypotenuse in the triangle defined by δx and δy ;
- 2 summing all such contributions between $x = a$ and $x = b$, and letting $\delta x \rightarrow 0$.



Length of a Curve - Example

Formula

$$\text{length of curve} = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

Example: Find the length of the curve $y = 3x + 2$ between $x = 1$ and $x = 5$.

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Area of a Surface of Revolution

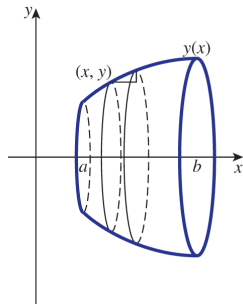
Given a curve with equation $y = f(x)$, then the area of the surface generated by rotating that part of the curve between the points $x = a$ and $x = b$ around the x axis through 360° is given by the formula

Formula

$$\text{area of surface} = \int_a^b 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

We can prove this formula by:

- ① selecting a small disc of the solid of revolution;
- ② calculating the length of its circumference;
- ③ approximating the surface area of this disc using Pythagora's for the length of the curve;
- ④ summing all contributions to the surface area between $x = a$ and $x = b$, and letting $\delta x \rightarrow 0$.



Area of a Surface of Revolution - Example

Formula

$$\text{area of surface} = \int_a^b 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

Example: Find the area of the surface generated when the part of the curve $y = x^3$ between $x = 0$ and $x = 4$ is rotated around the x axis.

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Mean Value and Root-Mean-Square Value

Physical quantities often vary with time (e.g. velocity, voltage, blood pressure) and we may wish to know the **average value** of such quantities over some particular time interval.

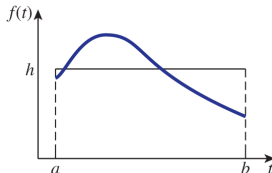
The average value of a time-varying function is defined in terms of an integral.

An associated quantity is the **root-mean-square (r.m.s.) value** of a function. For example, the r.m.s. value of an electrical current is used in the calculation of the power dissipated by a resistor.

Mean Value of a Function

Suppose a function $f(t)$ is defined on the interval $a \leq x \leq b$. The area, A , under the graph of $f(t)$ is given by the integral

$$A = \int_a^b f(t) dt$$



We can draw a rectangle with base spanning the interval $a \leq x \leq b$ and which has area A . The height of the rectangle, h , is defined as the **mean value** of the function across the interval $a \leq x \leq b$. It can be calculated using

$$h(b-a) = A = \int_a^b f(t) dt$$

Formula

$$\text{average value} = h = \frac{1}{b-a} \int_a^b f(t) dt$$

It is also referred to as the **average value**.

Mean Value of a Function - Example

Formula

$$\text{average value} = h = \frac{1}{b-a} \int_a^b f(t) dt$$

Example: Find the average value of $f(t) = t^2$ across the interval $1 \leq t \leq 3$.

Root-Mean-Square Value of a Function

If $f(t)$ is defined on the interval $a \leq x \leq b$, the **root-mean-square (r.m.s.) value** is given by the following formula:

Formula

$$\text{r.m.s value} = \sqrt{\frac{1}{b-a} \int_a^b [f(t)]^2 dt}$$

Note that

$$\frac{1}{b-a} \int_a^b [f(t)]^2 dt$$

is the mean value of $[f(t)]^2$ over the interval from a to b , that is the **mean square value**.

Example: Find the r.m.s. value of $f(t) = t^2$ across the interval $1 \leq t \leq 3$.

What We Learnt Today

- To understand how to use the interpretation of the definite integral as the limit of a sum for calculating geometric and physical quantities;
- To be able to calculate volumes of revolution, lengths of curves, and areas of surfaces of revolution;
- To be able to calculate centres of mass, moments of inertia, mean and root-mean-square values.